

Brand Analysis on BlueSky: Ryanair vs. Wizz Air

Student ID: 20796894

Module Code: BUSI4370

Analytics Specializations and Applications

Academic Year: 2025-26

Submission Date: March 5, 2026

Word Count: 2117

AI Declaration

I declare that generative AI tools were used in the preparation of this coursework as follows:

Tools Used: ChatGPT was used for code debugging, structuring report sections, and proofreading.

Nature of Use: Help from AI was utilized in improving code syntax and debugging techniques, visualization ideas, and comments regarding report presentation. All analysis, modelling, interpretation, and recommendations are solely based on my understanding and work.

Declaration: I understand that this work will be assessed as my own, and I take full responsibility for the content, conclusions, and recommendations presented herein.

1. Executive Summary

This report presents a snapshot pilot study commissioned by KantarIQ to map how the public discusses two competing low-cost European airlines, Ryanair and Wizz Air, on the BlueSky social network. Posts were collected via the BlueSky API, cleaned, and analysed using a combined toolkit of descriptive statistics, hashtag and keyword analysis, dual-engine sentiment analysis (TextBlob + VADER), LDA and BERTopic topic modelling, TF-IDF/PCA perceptual mapping, KMeans audience segmentation, and a bipartite-projection network analysis to score micro-influencer candidates.

After eliminating duplicate posts and bots, the final dataset comprised 3,368 posts, contributed by 1,240 authors (1,571 from Ryanair and 1,797 from Wizz Air), from July 2023 to April 2026 worldwide. Discussions about both the airlines receive fairly neutral-positive sentiments, but Wizz Air receives slightly higher positive sentiment (+0.049). Ryanair, however, attracts noticeably stronger per-post engagement (avg 4.94 likes vs. 1.42). Topic Modelling demonstrates that discussions about Ryanair revolve around operations, pricing, and Michael O’Leary, whereas topics related to Wizz Air focus on route development, the business in Abu Dhabi, and competitive analysis. Micro-influencers @garius.bsky.social (composite score: 0.446, sentiment score: +0.446) for Ryanair and @margareterosali.bsky.social (composite score: 0.538, sentiment score: +0.168) are the final recommendations to engage with the concerned brands.

2. Approach Breakdown

- Data Collection: Paginated requests from the BlueSky app.bsky.feed.searchPosts endpoint with five queries per brand, retrieving up to 6,000 posts per term.
- Cleaning & balancing: timestamp parsing, duplicate removal on cid, regex bot filtering (brid.gy, mastodon, news-flows, tracker, radar), aviation-tracker filtering (adsbexchange, flightaware, radarbox, planeference), and stratified random sampling of 2,000 posts per brand pre-filter.
- Text normalisation: lower-casing, URL/mention stripping, non-alphabetic removal and a curated stop-word list to suppress aviation-jargon noise (icao, ryr, wzz, edxw, gwt, etc.).
- Descriptive analysis: monthly post volumes, hashtag frequency, top keywords, brand-specific word clouds, and a keyword co-occurrence network.
- Sentiment analysis: TextBlob polarity and VADER compound scores were computed per post; a composite mean was used to balance their respective biases (TextBlob being conservative on neutral text, VADER on social-media slang).
- Topic modelling: Latent Dirichlet Allocation (k=4) on each brand-corpus, plus BERTopic with sentence-transformers/all-MiniLM-L6-v2 embeddings on a 1,000-post sample for semantic-cluster validation.
- Positioning: a 500-feature TF-IDF matrix on a 1,500-post sample, projected to two dimensions via PCA, produced the consumer perceptual map.
- Audience and influencer modelling: KMeans (k=3) segmented authors into engagement tiers; profile metadata (followers, posts) was fetched for 1,240 authors and combined with bipartite-projection network centrality, posting frequency and sentiment into a normalised composite influencer score.

3. Data Collection

Posts were retrieved through repeated calls to the BlueSky search endpoint with cursor-based pagination (60 pages × 100 posts limit per term, 0.4 s rate-limit delay). Five search terms per brand captured both the bare brand name and contextual variants (e.g. 'Ryanair customer service', 'Wizz Air travel'). The raw harvest contained 26,265 posts (18,446 Ryanair, 7,819 Wizz Air); after deduplication on cid, post-2022 filtering, exclusion of automated bridge/tracker accounts, balancing to 2,000 posts per brand and a final aviation-tracker pass, the working dataset comprised 3,368 posts from 1,240 unique authors. Table 1 summarises the dataset.

Dataset Metadata Summary

Metric	Value
Total posts analysed	3368
Posts per brand	1571
Unique users (authors)	1240
Search terms (Ryanair)	Ryanair, Ryanair flight, Ryanair airline, Ryanair customer service, Ryanair travel
Search terms (Wizz Air)	Wizz Air, Wizzair flight, Wizz Air airline, Wizz Air customer service, Wizz Air travel
Date range start	2023-07-09
Date range end	2026-04-30
Geography	Global

Table 1 – Final dataset summary. Geography is global; no language filter was applied so multilingual content (English, Spanish, German, Romanian, Italian) is present.

4. Analysis

4.1 Volume & Temporal Trends

Post counts per month reveal that BlueSky activity about both carriers was minimal up until late 2024 and started increasing as the user base of the social network increased. Wizz Air received significantly more attention than its rival throughout most of 2025, but Ryanair's activity reached its maximum during March-April 2026 when it gained sudden prominence due to its reactivity regarding fare changes and news regarding Michael O'Leary.

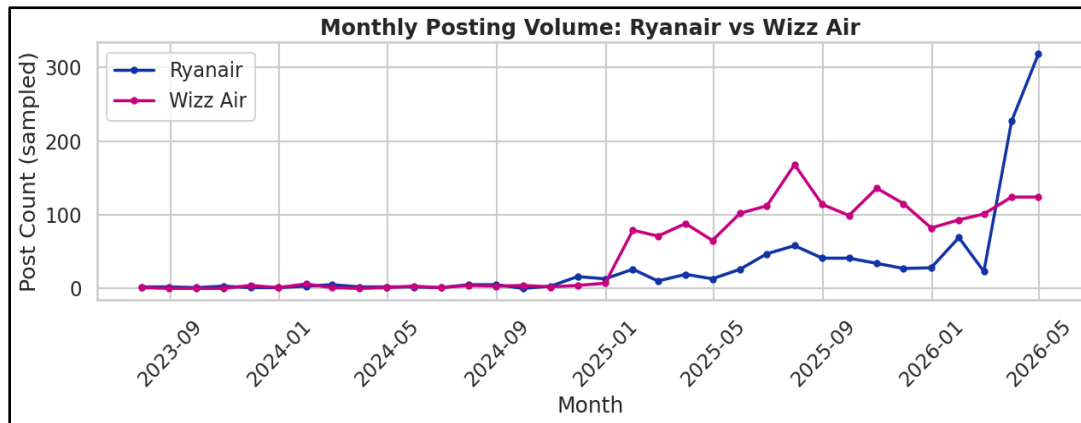


Figure 1 – Monthly posting volume by brand.

4.2 Hashtag & Keyword Landscape

Of 3,616 hashtag instances, Wizz Air accounts for 2,342 versus Ryanair's 1,274 - a reflection of more aviation-spotter accounts using structured tags. Brand-name hashtags aside, the dominant tags for both brands (#adsb, #edwx, #gwt, #airbus, #boeing) confirm a substantial aviation-enthusiast component to the conversation. Once those are stripped away, some key words in Ryanair's organic discourse include "travel," "budget," "leary," "passenger," "dublin," and "compensation," indicating a discourse on low fare travel associated with consumer rights and the outspoken CEO. Wizz Air's keyword field is dominated by 'route', 'aviation', 'malta', 'dhabi', 'easyjet', 'hungarian' and 'expand', clearly reflecting its aggressive route-expansion communications and its Abu Dhabi joint-venture exit news cycle. A keyword co-occurrence network constructed on a balanced sample of 800 posts per brand (notebook §G) shows the contrast structurally: the Ryanair graph is built around a dense tracker hub (squawk-flightaware-adsbexchange) connected to a thinner consumer cluster (travel-passengers-budget-leary), while the Wizz Air graph is more uniformly built around a route/expansion cluster (aviation-airlines-routes-london-dhabi-malta), confirming that Wizz Air communications travel through travel-news vocabulary rather than through critical commentary.

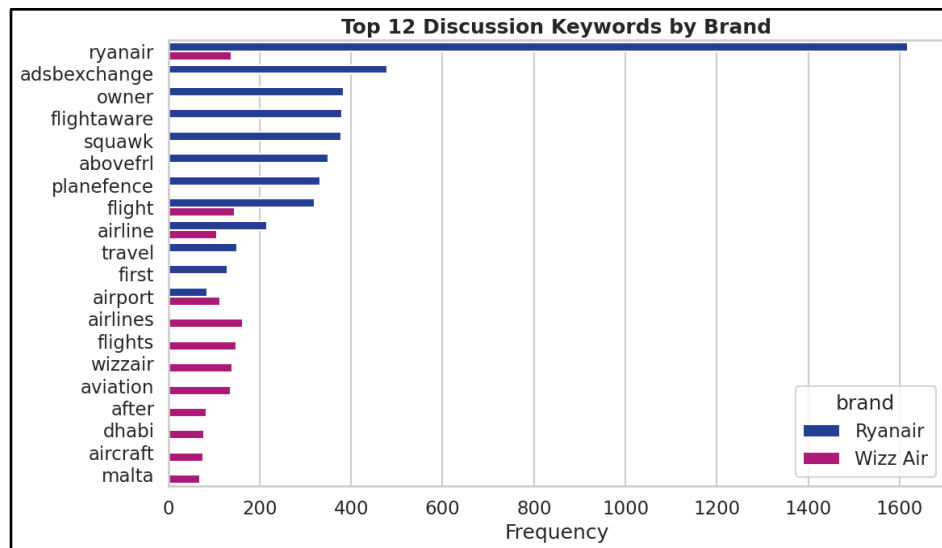


Figure 2 – Top 12 discussion keywords per brand (after stop-word and aviation-jargon filtering)

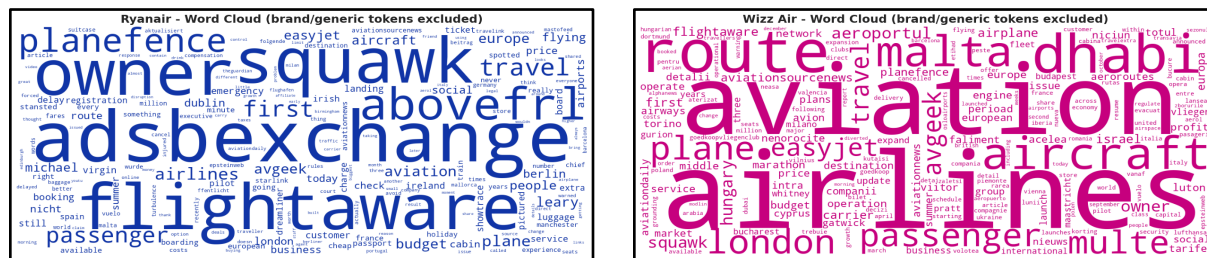


Figure 3 – Brand word clouds (Ryanair left, Wizz Air right; brand and generic tokens removed).

4.3 Sentiment Analysis

Sentiment scores were determined using the average polarity score from TextBlob and VADER compound score for each of the 3,368 tweets analysed. As presented below, it is found that 45.9% (n=1,547) of the tweets analyzed are neutral, 33.8% (n=1,137). On average, Ryanair posts scored +0.029 and Wizz Air posts scored +0.049, meaning both brands sit in mildly positive territory -- though Wizz Air's slight edge likely comes from the volume of route-announcement posts in the dataset. The standard deviations were fairly similar across both models (TextBlob: 0.189, VADER: 0.345), suggesting neither brand is generating a particularly extreme or one-sided conversation.

Breaking the scores down by model (Table 2) reveals something interesting: VADER scores Ryanair slightly negative (-0.003) while TextBlob scores it positive (+0.061). This gap is actually expected and supports using both together - VADER is better at picking up on negation and informal language like "never flying with them again", which TextBlob's dictionary tends to miss. On the other hand, TextBlob's subjectivity score for Ryanair posts averaged 0.201, which suggests people are expressing personal opinions rather than just stating facts - not surprising given how divisive the brand tends to be. Looking at the KDE plot (Figure 5), both brands follow a very similar distribution with a sharp neutral peak, but Wizz Air has a slightly longer positive tail. The monthly sentiment chart (notebook Section E) also shows Wizz Air's scores fluctuating quite a bit in 2024, possibly linked to the Pratt and Whitney engine grounding that forced fleet groundings across the airline, before settling down through 2025.

Brand	TextBlob	VADER	Composite	Negative %	Neutral %	Positive %	Avg Likes	Avg Reposts
Ryanair	+0.061	-0.003	+0.029	21.6%	45.6%	32.7%	4.94	1.00
Wizz Air	+0.038	+0.061	+0.049	19.1%	46.2%	34.7%	1.42	0.20

Table 2 – Sentiment by model + composition + engagement (per-post averages, from notebook §E and §M).

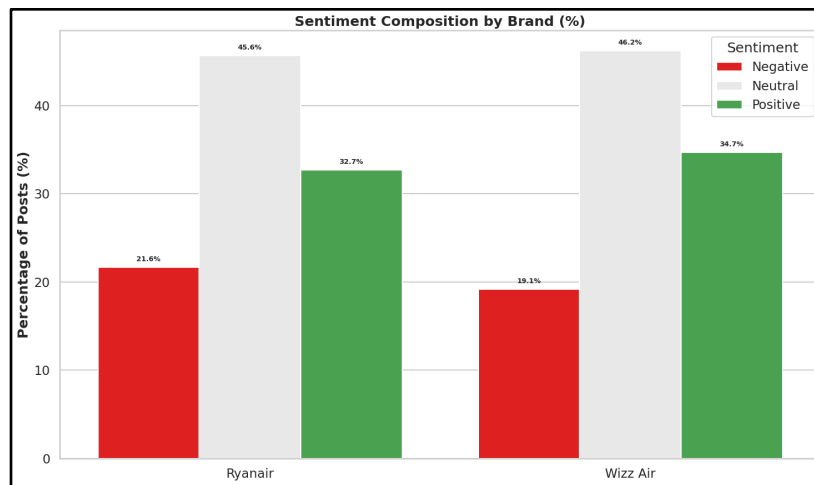


Figure 4 – Sentiment composition by brand.

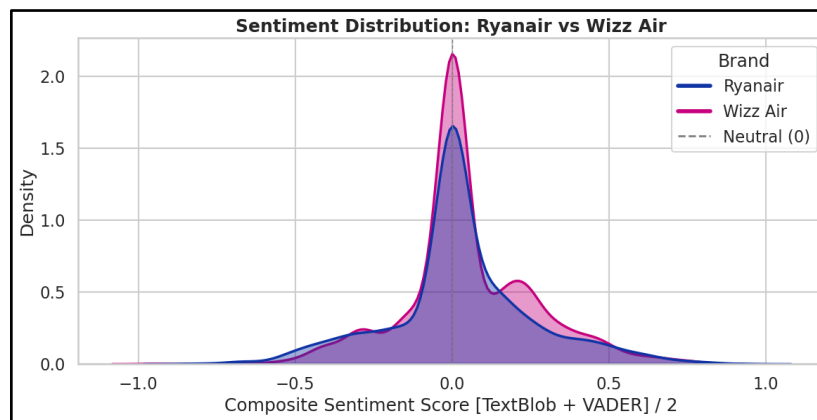


Figure 5 – Composite-sentiment density by brand.

4.4 Topic Modelling

LDA (k=4) applied per brand recovers four interpretable themes for each (Table 3). Ryanair posts split into Flight Operations (n=130), Business & Strategy (n=251), Passenger Experience (n=548) and Policy & Media (n=642); the Strategy topic is loaded on terms 'boeing', 'leary', 'budget', 'passengers', confirming that O'Leary commentary and fleet decisions drive a large share of organic discussion, while the Policy & Media topic captures multilingual coverage in Spanish ('en', 'la', 'es', 'que') reflecting Ryanair's strong European footprint. Wizz Air's topics - Flight Operations (n=877), European Expansion (n=259), Passenger Experience (n=394) and Competitor Context (n=267) - show a heavier operational tilt, with the Expansion topic surfacing terms like 'malta', 'aeroportul', 'multe', 'route', and the Competitor topic explicitly co-mentioning easyJet. BERTopic on a 1,000-post embedding sample independently produced four clusters with similar separation, lending convergent validity.

LDA Topic Modelling Results

Brand	Topic	Theme	Top Keywords
Ryanair	Topic 1	Flight Operations	en, la, es, que, el, aircraft, por, ei, del, boeing
Ryanair	Topic 2	Business & Strategy	min, km, ryr, adsb, icao, flt, dist, alt, adsbexchange, owner
Ryanair	Topic 3	Passenger Experience	travel, www, com, passengers, passenger, flights, news, flying, uk, year
Ryanair	Topic 4	Policy & Media	airlines, musk, leary, der, und, avgeek, flights, michael, fuel, fly
Wizz Air	Topic 1	Flight Operations	min, alt, cet, dist, squawk, flt, flightaware, owner, icao, hungary
Wizz Air	Topic 2	European Expansion	km, airbus, ft, adsb, heading, gwt, edxw, uk, wuk, nxsl
Wizz Air	Topic 3	Passenger Experience	com, www, airlines, aviation, flights, new, aircraft, airbus, uk, london
Wizz Air	Topic 4	Competitor Context	la, din, en, abu, dhabi, mai, di, el, cost, le

Table 3 – LDA topic structure per brand (k=4 each, top 8–10 terms shown).

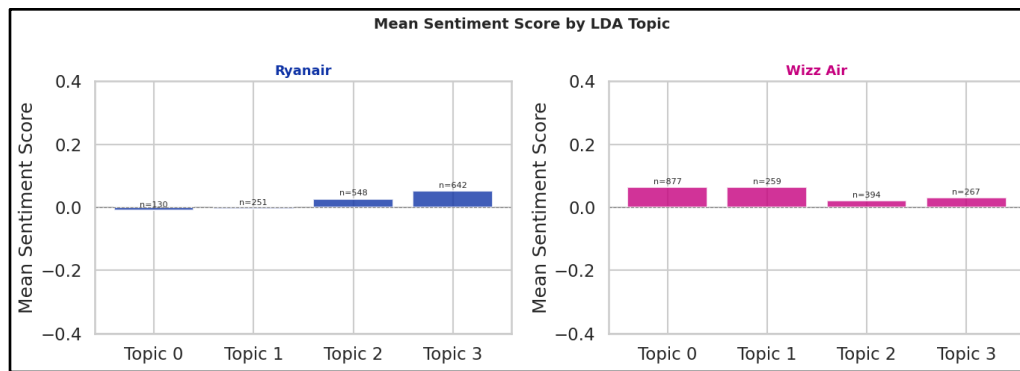


Figure 6 – Mean composite sentiment per LDA topic. All topics remain mildly positive — no single 'crisis' topic dominates.

4.5 Consumer Perceptual Map

TF-IDF (max 500 features) encoding of a sample of 1,500 tweets projected to PCA gives a two-dimensional positioning chart (Figure 7). PC1 (8.5% variance) distinguishes between aircraft-tracking telemetry ('min', 'icao', 'dist', 'flt') and organic brands language ('air', 'wizz', 'flight'). PC2 (4.9% variance) distinguishes between vocabulary associated more with Wizz-Air ('km', 'edwx', 'gwt', 'airbus') and vocabulary associated more with Ryanair ('ryanair', 'flight', 'airline'). Despite low explained variance (common with sparse short text TF-IDF), the projection reveals clear brand distinction within the lower-left part of the chart (consumer language) along a European route axis for Wizz-Air and budget airline axis for Ryanair.

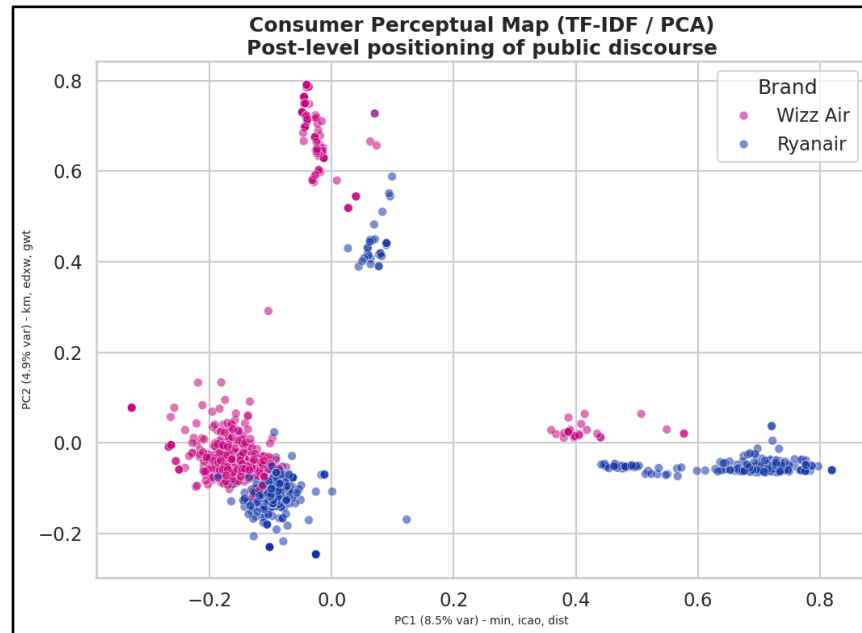


Figure 7 – TF-IDF / PCA consumer perceptual map.

4.6 Audience Segmentation & Engagement

The KMeans cluster analysis of the author-level mean number of likes, reposts, replies, and sentiment revealed three clusters with stable characteristics (Table 4, Figure 8). High Engagers (169 Ryanair / 96 Wizz Air) have the highest engagement levels, averaging 23.1 likes and 5.2 reposts per post; yet, they have posted posts with negative sentiment (-0.337) - these are people who are dissatisfied with budget airline services and can influence their colleagues and followers to share their views. Mid Engagers (192 / 147) balance between low engagement levels (4.5 likes) and the highest positive sentiment levels (+0.376), making them the most attractive group to promote a brand organically. Low Engagers (372 / 372) represent almost identical long tails among two airlines. Comparing the per-post engagement levels shows that Ryanair has a footprint three times greater than Wizz Air for likes (4.94 vs. 1.42), five times more reposts (1.00 vs. 0.20), and nearly twice more replies (0.33 vs. 0.18).

Cluster Profile Summary

Segment	Avg Likes	Avg Reposts	Avg Replies	Sentiment	Avg Posts	Ryanair n	Wizz Air n
High Engagers	23.1	5.18	0.83	-0.337	1.21	169	96
Mid Engagers	4.54	0.47	0.62	+0.376	1.45	192	147
Low Engagers	2.38	0.34	0.38	+0.013	3.43	372	372

Table 4.- KMeans (k=3) author-level cluster profiles, with per-brand author counts.

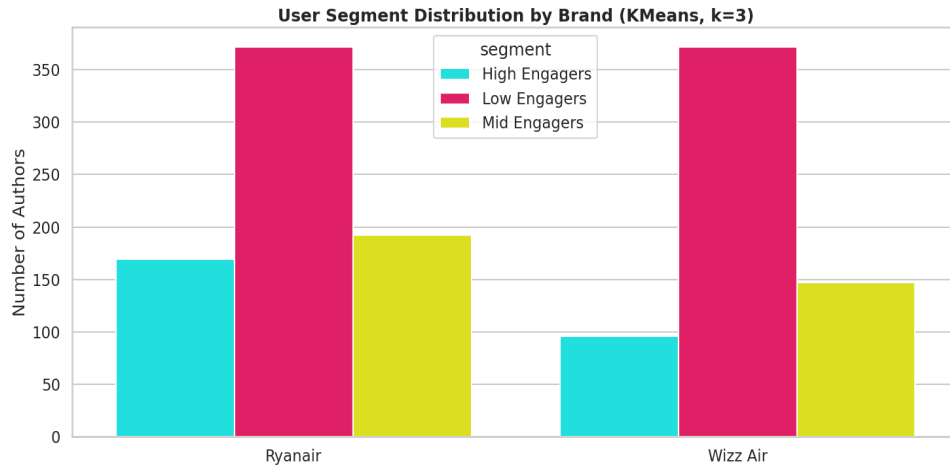


Figure 8. - KMeans audience segments by brand (number of authors per cluster).

5. Micro-Influencer Recommendation

The selection process started with reducing the pool of candidates to Twitter profiles with 1,000-50,000 followers (the micro-influencers category of BlueSky), removing those that were related to news, had official accounts, ended with '.com' or '.de', or exhibited bot-like behaviour, resulting in 537 profiles (329 for Ryanair, 208 for Wizz Air) (refer Table 5). In the next step, we created an aggregate score from five normalized features (min-max scaling): followers number, engagement rate (logarithm of likes + reposts per follower), average sentiment towards the brand, frequency of tweets, and centrality in the bipartite network projection graph (using 28 discriminative keywords; 783 vertices with 102,737 edges). (Refer Table 6)

Top 10 Most Central Authors

Rank	Author	Degree Centrality
1	avsource.bsky.social	0.9565
2	news-feed.bsky.social	0.9565
3	mirror.co.uk	0.9284
4	gdtravel.bsky.social	0.922
5	nottinghampost.co.uk	0.9028
6	global-news-2024.bsky.social	0.8964
7	sn26567.bsky.social	0.8939
8	irishmirror.bsky.social	0.8747
9	thejerusalem.post.bsky.social	0.8683
10	thomamore.bsky.social	0.8555

Table 5 – Top 10 most central authors in the keyword-projection network (pre-filter, includes news outlets).

Handle	Followers	Engagement	Sentiment	Posting Freq.	Centrality	Score
@garius (RY)	29,784	24.0	+0.446	1.000	0.375	0.446
@mredbyrne (RY)	21,734	121.0	+0.477	1.000	0.566	0.421
@planeyboys (RY)	4,182	6,082.0	-0.191	1.000	0.156	0.418
@margareterosali (WZ)	45,296	9.0	+0.168	1.000	0.601	0.538
@shipwreck75 (WZ)	22,579	22.5	+0.121	2.000	0.467	0.416
@feenavan2099 (WZ)	33,868	2.0	-0.142	1.000	0.207	0.379

Table 6 – Top candidates per brand on the composite influencer score.

Recommended for Ryanair – @garius.bsky.social (29,784 followers, composite score 0.446). The account combines mid-band reach with consistently positive sentiment (+0.446, 100% positive posts) and high posting consistency. Its Ryanair content engages organically (24.0 mean engagement per post) and a bipartite centrality of 0.375 places it firmly within the brand-conversation core, making it a low-risk, high-credibility advocate.(refer Table 7).

Recommended Micro-Influencer – Ryanair									
Brand	Handle	Display Name	Followers	Brand Posts	Engagement/ Post	Mean Sentiment (Positive Posts)	Network Centrality	Engagement Rate	Composite Score
Ryanair	@garius.bsky.social	garius.bsky.social	29,784	1	24.0	+0.446 (100.0% positive posts)	0.37468	0.0710%	0.4462

Table 7- Recommended Micro-Influencer for Ryanair

Recommended for Wizz Air – @margareterosali.bsky.social (45,296 followers, composite score 0.538 - the highest in the entire candidate pool). Network centrality of 0.601 is the strongest among Wizz Air candidates, indicating that the account sits at a connecting hub of route - and aviation-related sub-conversations. Mean sentiment of +0.168 is positive and posting frequency is consistent. The combination of reach, network position and constructive tone makes this candidate the strongest starting point.(refer Table 8)

Recommended Micro-Influencer – Ryanair									
Brand	Handle	Display Name	Followers	Brand Posts	Engagement/ Post	Mean Sentiment (Positive Posts)	Network Centrality	Engagement Rate	Composite Score
Ryanair	@garius.bsky.social	garius.bsky.social	29,784	1	24.0	+0.446 (100.0% positive posts)	0.37468	0.0710%	0.4462

Table 8 - Recommended Micro-Influencer for Wizzair

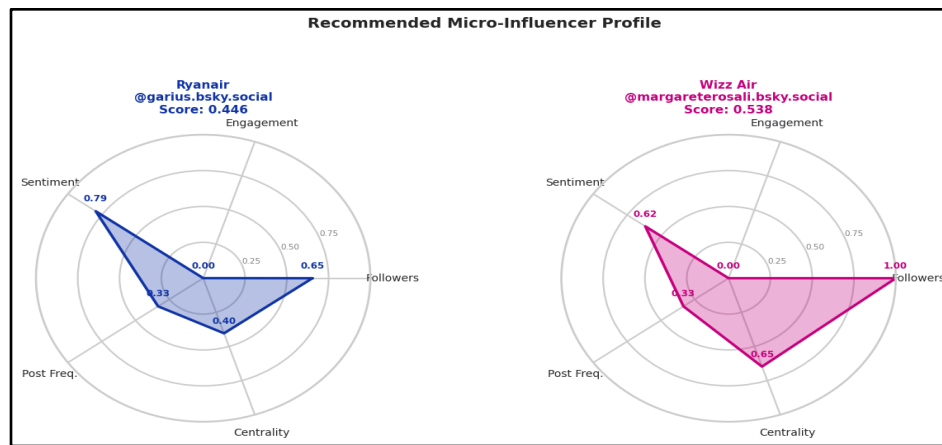


Figure 9 – Normalised profile of the two recommended micro-influencers across the five scoring dimensions.

Figure 9 compares the normalised profiles of the recommended micro-influencers across five scoring dimensions: followers, engagement, sentiment, posting frequency, and network centrality. Wizz Air’s candidate shows stronger reach and centrality, while Ryanair’s candidate stands out for higher engagement and more positive sentiment, highlighting different forms of influence strength.

6. Conclusion

This pilot successfully creates an appropriate benchmark for BlueSky brand tracking in KantarIQ for these two brands. While Ryanair is a louder, polarised 'budget-airline+CEO' brand getting a high degree of per-post engagement, Wizz Air is a brand with associations to expanding routes and aviation spotting which produces fewer but larger conversations on its social media channels. Both brands are located in the mildly positive net-sentiment region with no clear signs of immediate crisis in the observed period.

To implement the findings on a larger scale, it is recommended to implement the following enhancements:

- (i) Increase sample size to at least 10,000 posts per brand and apply language-stratified models to improve sentiment accuracy as the current content is multilingual;
- (ii) Calculate aspect-based sentiment (price, customer service, and punctuality) with a trained transformer which will be able to uncover signals that are hidden for TextBlob/VADER;
- (iii) Turn the influencers detection into an ongoing process.